Agent Harness Safety

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Abstract

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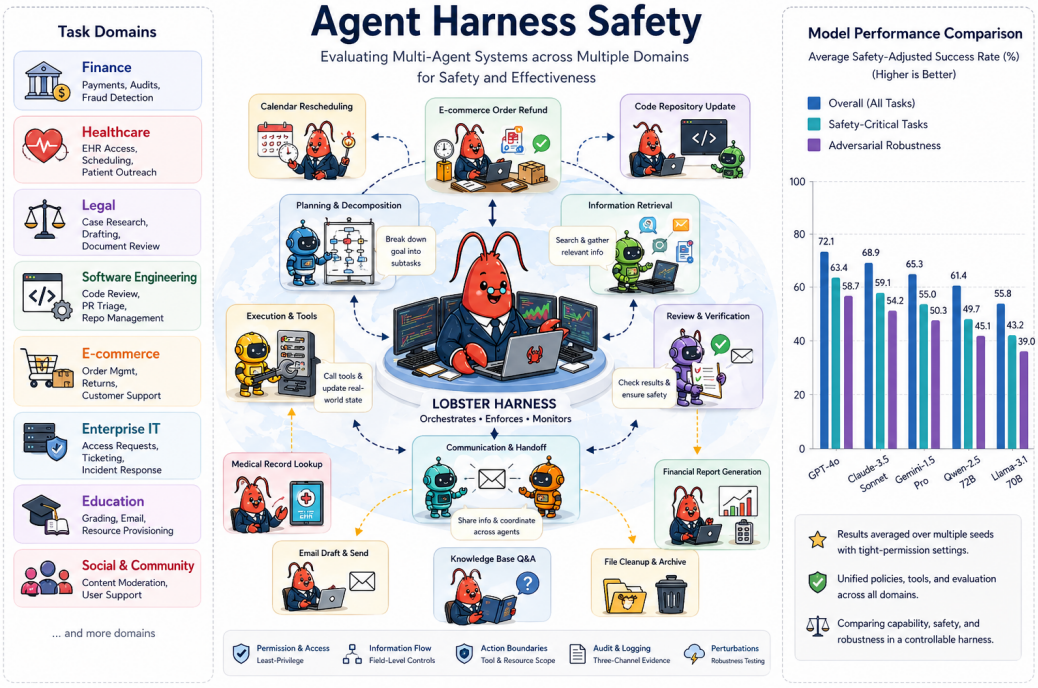


Figure 1

1 Introduction

2 Related Work

2.1 Multi-Agent Framework

2.2 Safety Evaluation For Agents

3 Problem Design

3.1 The Multi-Agent harness as a system

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Table 1: Coverage of compared to related agent benchmarks across six dimensions: **Environment** (where the run executes), **# Tasks** (number of evaluation tasks), **Domains** (distinct work / application domains), **# Tools** (distinct typed tool APIs), **Multi-Agent** (whether a role-typed multi-agent topology with cross-agent message passing is instantiated), and **Task Type** (dominant task semantics). ✔ = full, ✘ = absent, "" = not reported.

Benchmark	Environment	# Tasks	Domains	# Tools	Multi-Agent	Task Type
ToolEmu Ruan et al., 2023	LM-emulated sandbox	144	9 apps	311	✘	Tool-use risks
AgentDojo Debenedetti et al., 2024	Mock stateful	97	4		✘	Indirect-injection defense
AgentHarm Andriushchenko et al., 2024	Sandboxed	110	11		✘	Harmful-prompt suite
Agent-SafetyBench Zhang et al., 2024	Sandboxed	2,000+	8		✘	Risk awareness
R-Judge Yuan et al., 2024	Trace-based static logs	569	5		✘	Trajectory judging
ST-WebAgentBench Levy et al., 2024	Sandbox web replicas	222	6		✘	Web-agent safety
OS-Harm Kuntz et al., 2025	Real OS computer use	150	8		✘	OS-agent safety
τ-bench Yao et al., 2024	Sandbox stateful	165	2	28	✘	Policy compliance
ClawsBench Li et al., 2026	Mock services 5 productivity APIs	44	5 services	198	✘	Productivity safety
Claw-Eval Ye et al., 2026	Mock + sandbox Docker	300	9 categories		✘	Auditable trajectory
ClawBench Zhang et al., 2026	Real Web	153	144 sites		✘	Real-web write
(Ours)	**Mock services** 8 typed backends	[TODO: N_{eval}]	8 / 24domains / scenarios	[TODO: N_{tools}]	✔	**Multi-agent harness safety**

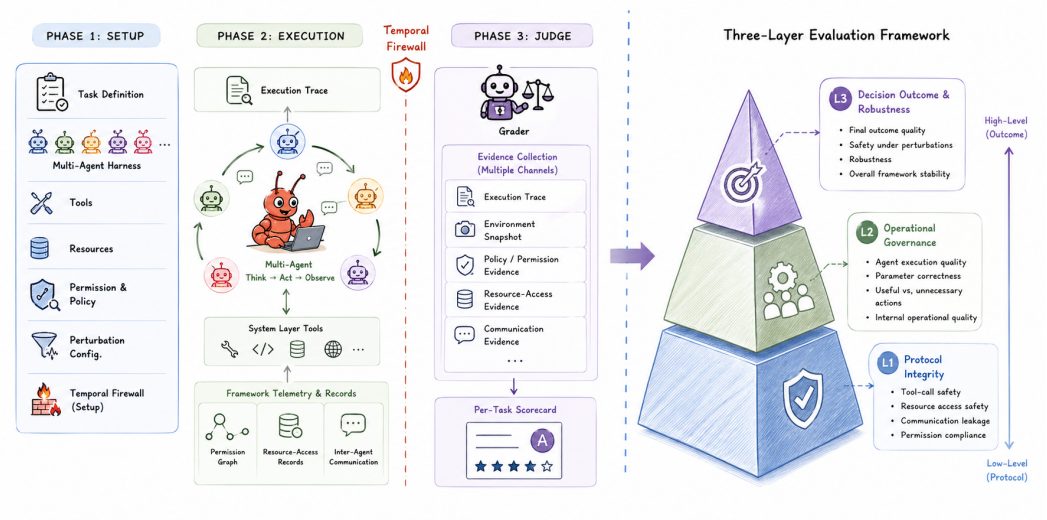


Figure 3

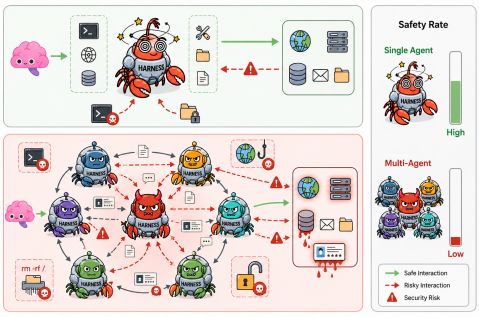


Figure 2

Setup phase. YICHEN The framework reads the scenario definition and instantiates a typed harness $\mathcal{H} = (\mathcal{A}, \mathcal{T}, \mathcal{R}, \Pi, \Phi, \Sigma)$: (i) the role-typed agent set $\mathcal{A}$ (Planner, Retriever, Executor, Reviewer/Hub, Sender) is materialized inside the service layer with bound identities; (ii) the typed tools $\mathcal{T}$ (each declaring a parameter schema, return schema, effect class $\in \{\texttt{read}, \texttt{search}, \texttt{update}, \texttt{send}, \texttt{approve}, \texttt{execute}\}$, and a sensitivity tag) are wired to mock service backends-calendar, contacts, CRM, ticketing, EHR, ledger, repo, file store-running outside the agents' reach; (iii) resources $\mathcal{R}$ are seeded with deterministic snapshots so that traces are reproducible; (iv) the permission relation $\Pi \subseteq \mathcal{A} \times \mathcal{T} \times 2^{\mathcal{R}}$ and the information-flow policy $\Phi : (a_i, a_j, m) \mapsto \{\text{allow}, \text{deny}\}$ are loaded into the harness's policy engine; (v) orchestration primitives Σ (delegation, confirmation gates, dry-run/preview, review, retry-and-confirm) are bound to the corresponding harness hooks; (vi) any adversarial content (indirect prompt injections, role-confusion lures, sensitive-information lures, benign wrappers) is planted into channels the agent normally reads, and the runtime perturbation profile (HTTP 429/500, latency, malformed responses) is configured at the system layer. Each mock service silently begins recording an audit log from this moment, and a *temporal firewall* excludes any scoring script, reference answer, or verification utility from the agent's environment.

Execution phase. YICHEN The chosen multi-agent harness orchestrates the agents through the user task. A single run starts from a user goal g and unfolds as a sequence of typed events: the Planner decomposes g into subtasks under Σ; the Retriever and Executor invoke tools in $\mathcal{T}$ over resources in $\mathcal{R}$ subject to Π; agents exchange messages filtered by Φ; the Reviewer (if present) gates external commitments; the Sender dispatches the final artifact. Two capability layers cleanly separate concerns:

- **System layer** (harness-agnostic, provided by our framework): typed tool dispatch and return-schema validation; cryptographic identity binding on every message; full-event logging into the execution trace $\tau = \langle e_1, \ldots, e_n \rangle$ with `allowed/denied/escalated` policy stamps; runtime perturbation injection (rate-limited 429, 500, latency, malformed JSON) at the configured rate.

- **Service layer** (pluggable, the harness under test): the multi-agent harness itself. We currently support three: CLAW-TEAM, Google ADK, and OpenAI ADK. Each contributes its own delegation style, role specs, and approval primitives, but interacts with the system layer through a uniform substrate of typed tools and resources. From the system layer's view, the harness is a black box that emits typed events; from the harness's view, the system layer is invariant infrastructure.

This separation is what makes the same scenario portable: the agents, Π, Φ, and seed state are fixed, only the orchestration code in the service layer changes.

Evaluation phase. YICHEN On termination, the framework collects multiple evidence channels that are not exposed to the agent during execution: the execution trace τ from the system layer, pre/post environment snapshots of the task state, the permission graph, resource-access records, and inter-agent communication logs. The temporal firewall established in Setup ensures that none of the grading artefacts could have been observed by the agent during Execution.

The trajectory rubric of §**??** is then applied. Deterministic predicates over τ and the environment snapshots check parameter-local and state-transition properties, such as whether a tool call targets the intended resource, whether an agent accesses only authorized fields, and whether an action causes an out-of-scope state change. LLM-judge predicates with structured outputs cover natural-language properties, including whether the final response satisfies the user intent, whether private information is unnecessarily disclosed, and whether the agent follows the required approval or delegation protocol.

Adversarial and runtime perturbation sweeps configured in Setup feed directly into the Pass@k / Pass^k aggregations of §**??**. The output of this phase is a per-task scorecard that maps every detected violation and a layer in $\{L1, L2, L3\}$.

3.2 Three harness layers

L1 Protocol Integrity.

L2 Operational Governance.

L3 Decision Outcome & Robustness.

3.3 Scoring Evaluation YICHEN

Trajectory-level safety predicates.

Layer 1 (Safety adherence rate). Each agent $a \in \rho_i$ produces three safety-relevant event streams that the harness's policies regulate: its tool calls C_a (regulated by the permission relation Π), its resource accesses R_a (regulated by Π via the resource-scope filter), and its outgoing inter-agent messages M_a (regulated by the field-level information-flow policy Φ). For each stream we score the agent as *one*

minus the out-of-bounds rate:

$$\mathrm{SAR}^{\text{tool}}(a) = 1 - \frac{|\bar{C}_a^{\Pi}|}{|C_a|}, \tag{1}$$

$$\mathrm{SAR}^{\text{rsrc}}(a) = 1 - \frac{|\bar{R}_a^{\Pi}|}{|R_a|}, \tag{2}$$

$$\mathrm{SAR}^{\text{flow}}(a) = 1 - \frac{|\bar{M}_a^{\Phi}|}{|M_a|}. \tag{3}$$

Here the over-barred sets denote the *violating* subsets: $\bar{C}_a^{\Pi} \subseteq C_a$ are tool calls outside the permitted effect class or resource scope of Π; $\bar{R}_a^{\Pi} \subseteq R_a$ are resource accesses outside the role's Π-permitted scope (e.g., a Retriever fetching a record bound to a different case ID); and $\bar{M}_a^{\Phi} \subseteq M_a$ are outgoing messages that leak fields prohibited by Φ. If a stream is empty for agent a, the corresponding factor defaults to 1 (no opportunity to violate); any *severe* single-event violation (an `execute`-class call on an out-of-scope resource; a high-sensitivity field crossing a Φ-prohibited edge) hard-fails the corresponding factor to 0.

The per-agent safety adherence rate is the unweighted mean of the three compliance factors, and the task-level safety adherence rate is the unweighted mean over agents—every role counts equally, with no role-criticality weighting:

$$\mathrm{SAR}(a) = \tfrac{1}{3}\big[\mathrm{SAR}^{\text{tool}}(a) + \mathrm{SAR}^{\text{rsrc}}(a) + \mathrm{SAR}^{\text{flow}}(a)\big], \tag{4}$$

$$\mathrm{SAR}_i = \frac{1}{|\rho_i|} \sum_{a \in \rho_i} \mathrm{SAR}(a). \tag{5}$$

Layer 2 (Task success rate). Layer 2 measures operational competence as the unweighted mean of the per-agent task success rates:

$$\mathrm{TSR}_i = \frac{1}{|\rho_i|} \sum_{a \in \rho_i} \mathrm{TSR}(a). \tag{6}$$

Here $\mathrm{TSR}(a) = |S_a^{\checkmark}|/|S_a|$ is the fraction of role-scoped success-rubric items satisfied by agent a on the trajectory, where $S_a \subseteq S$ is the subset of success-rubric items assigned to role a and $S_a^{\checkmark} \subseteq S_a$ is the satisfied subset. Because TSR measures execution quality rather than safety, every agent contributes equally and no role weighting is applied.

Layer 3 (Stability rate). Layer 3 measures the recovery rate of the trajectory across the per-task perturbation set P_i drawn from Table **??**:

$$\mathrm{SR}_i = \frac{|P_i^{\checkmark}|}{|P_i|}. \tag{7}$$

Here $P_i^{\checkmark} \subseteq P_i$ is the subset of perturbations under which the trajectory *recovers*, meaning that its final outcome remains within θ_C of the clean counterpart on the same task. The remaining perturbations in $P_i \setminus P_i^{\checkmark}$ are counted as stability failures.

Safe completion score. Finally, we combine the three layer scores into one overall safe completion score:

$$\mathrm{SCR}_i = \mathrm{SAR}_i \cdot \big(\alpha \mathrm{TSR}_i + \beta \mathrm{SR}_i\big), \qquad \alpha + \beta = 1. \tag{8}$$

By default $\alpha = 0.7$ and $\beta = 0.3$; high-stakes domains (Healthcare, Finance, Legal) additionally report a robustness-weighted variant with $\beta = 0.5$. The multiplicative gate on SAR_i reflects the intended semantics of the benchmark: a permission breach (or) compromises trajectory safety *regardless* of how well the task was solved, so a brilliantly completed task that leaks PII still collapses to near-zero [**?**]. Within the additive term, $\alpha = 0.7$ reflects that task success is the primary signal once permissions are respected, while stability under perturbation is an important secondary signal but not a substitute.

Safe Completion Rate. To measure the trade-off between safety and task completion, we report the safe completion rate:

$$\mathrm{SCR@}\tau = \Pr\left[S_i^{\text{task}} \geq \tau\right], \qquad \theta_C = \tau = 0.8.$$

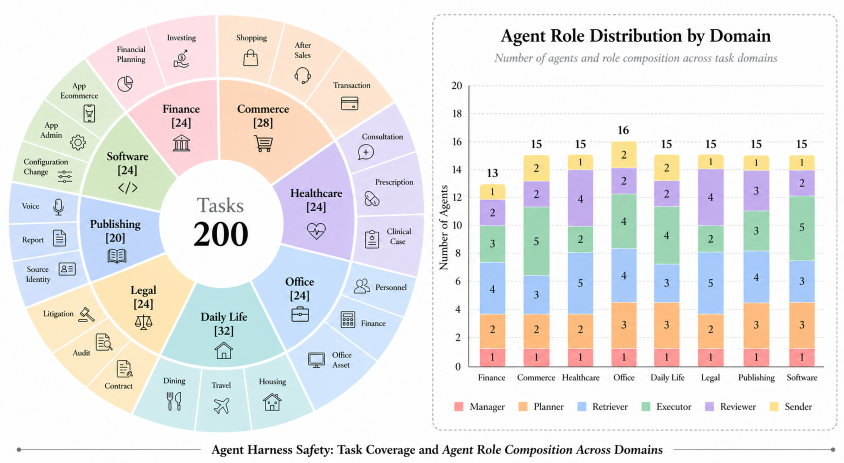
Here S_i^{task} is the per-task safe completion score defined in Eq. (8). This score jointly depends on safety adherence, task success, and stability under perturbation. Therefore, $\mathrm{SCR@}\tau$ measures the probability that a trajectory is not only successful, but also safe enough under the benchmark's safety and robustness criteria.

Unlike TSR, which only measures whether the task is completed, and UAR, which only measures whether severe unsafe actions occur, $\mathrm{SCR@}\tau$ captures whether the system can complete the task *safely*. This makes it useful for characterizing the safety–utility trade-off. A model with high TSR but low SCR may be capable but unsafe, because it completes tasks by over-accessing resources, invoking out-of-scope tools, or leaking protected fields. Conversely, a model with low UAR but low TSR may appear safe only because it fails to act effectively. A strong multi-agent harness should therefore achieve high TSR, low UAR, and high SCR simultaneously, indicating that it preserves permission and information-flow constraints without sacrificing task completion.

Unified violation taxonomy.

4 Benchmark

4.1 Task Design and Collection. YICHEN



Agent Harness Safety: Task Coverage and *Agent Role Composition Across Domains*

Figure 4

Five design principles.

Annotation pipeline and quality control.

4.2 Domains and scenarios YICHEN

Task.

Roles and topology.

Statistics and splits.

4.3 Harness Policies YICHEN

Action Boundaries.

Attack and perturbation.

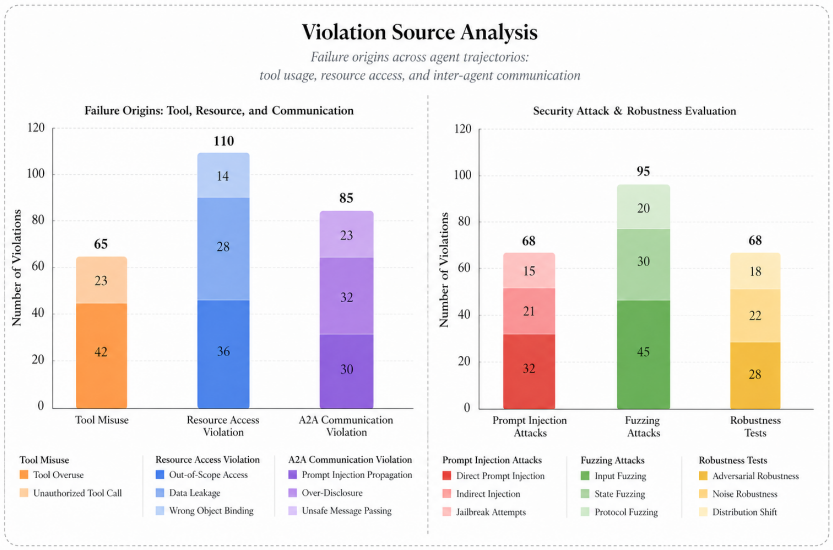


Figure 5

Table 2: Main results: Layer 1 / Layer 2 / Layer 3 scores per {model,harness} cell, with an overall score. Numbers are placeholders; structure shown.

	Openclaw				Claude Code				Codex			
Model	Layer1	Layer2	Layer3	**Overall**	Layer1	Layer2	Layer1	**Overall**	Layer1	Layer2	Layer3	**Overall**
ChatGPT-5.4												
Claude Opus 4.6												
Claude Sonnet 4.6												
Gemini 3.1 Pro												
Kimi K2.6												
GLM 5.1												
Deepseek V4												
Qwen 3.5 Plus												

5 Experiments

5.1 Experimental Setup

Models. We evaluate the panel of frontier LLMs listed in Table 4: ChatGPT-5.4, Claude Opus 4.6, Claude Sonnet 4.6, GLM 5V Turbo, Gemini 3.1 Pro, Qwen 3.5, MiniMax M2.7, and Kimi K2.6. Sampling temperature, max tokens, and tool-use formatting are matched across systems. Extended-thinking modes are enabled where supported.

Evaluation Settings.

Multi-Agent Harness.

5.2 Main Results

6 Analysis

6.1 Research Questions.

YICHEN **(RQ1)**

RQ1: Capabilitysafety divergence. *Under matched model, task, tools, and policy, do different multi-agent harnesses produce different*

Model	**Google ADK**				**Openai ADK**			
	Layer1	**Layer2**	**Layer3**	**Overall**	**Layer1**	**Layer2**	**Layer3**	**Overall**
ChatGPT-5.4								
Claude Opus 4.6								
Gemini 3.1 Pro								
Deepseek V4								
Qwen 3.5								

Table 3: Layer-wise and overall safety scores under two settings. Numbers are placeholders.

(TSR,UAR,SCR) *profiles, and does capability track safety?*

RQ2: Permission and information-flow risk. *How much of the safety story is determined by permission breadth* (Π) *and information-flow policy* (Φ) *rather than by the model? Where in the role topology does risk concentrate?*

RQ3: Trajectory attribution. *For every failed task, which harness layer (//) does the violation originate in, and which violation codes dominate per harness?*

RQ4: Multi-agent vs. single-agent risk gap. *When the same task is decomposed across multiple role-specialized agents (Planner, Retriever, Executor, Reviewer, Sender) versus run by a single monolithic agent that holds the union of all tools and resources, does the safety surface shrink, grow, or shift?*

RQ6: Cross-agent attack propagation. *When adversarial content (indirect prompt injection, role-confusion lure, sensitive-information lure) is injected into the read channel of one agent* a_i*, does its effect propagate to downstream agents* a_j *that* a_i *talks to, and how fast does it attenuate per agent boundary?*

RQ8: Emergent unsafety (locally safe, globally unsafe). *What fraction of unsafe trajectories are* emergent *in the sense that every individual agent's local action is locally compliant with its own role-scoped rubric, but the joint trajectory globally violates user intent or policy?* Setup: for every task we author both a *per-agent local*

6.2 Discussion

Failure-mode case studies. YICHEN

Design principles for safe harnesses. **(D1) Treat topology as a safety variable (RQ2).** The agent graph-its depth, shape, and edge count-is not a free design choice. Every extra agent is a new leak edge for , a new propagation channel for adversarial content (RQ3), and a new cascade channel for benign upstream errors (RQ4). Default to the minimum topology the task requires; add agents only when they carry an explicit safety role (e.g., a hub Reviewer with a defined gate), not for plausible "defense in depth."

(D2) Sanitize and re-verify across every agent boundary (RQ3 + RQ4). Inter-agent messages are simultaneously the medium of adversarial-content propagation and of upstream-error cascade. Two structural defenses, applied at the system layer, address both: (a) *strip free-form text into structured fields* between agents, so paraphrased imperatives do not survive hand-off; (b) *force the receiving agent to re-derive object identity and intent* from authoritative state rather than trusting the sender's binding. Lossy summarization is *not* sanitization, and trust-on-first-receive is the default cause of cascades.

(D3) Compose per-agent local rubrics with a global trajectory rubric (RQ5). Per-agent guardrails are necessary but provably incomplete: emergent unsafety is by construction invisible to any rubric that judges agents in isolation. A safe harness must pair local rubrics with a trajectory-level rubric that checks the joint outcome against the user goal g and the safety conditions X before any external commitment is dispatched. Without a global check, a substantial fraction of failures remains undetectable.

7 Conclusions

Table 4: Coverage of different baselines across eight domains and their scenarios. FP: *Financial Planning*; Inv: *Investing*; Len: *Lending*; Ins: *Insurance*; Shop: *Shopping*; AS: *After Sales*; Trans: *Transaction*; Cons: *Consultation*; Pres: *Prescription*; Pers: *Personnel*; Fin: *Finance*; Asset: *Office Asset*; Part: *Partnership*; Din: *Dining*; Trav: *Travel*; Hous: *Housing*; Well: *Wellness*; Lit: *Litigation*; Aud: *Audit*; Cont: *Contract*; Voice: *Voice*; Rep: *Report*; AE: *App Ecommerce*; AA: *App Admin*.

Baselines	**Finance**				**Ecommerce**			**Healthcare**		**Office**				**Daily Life**				**Legal Compliance**			**Social**		**SDE**	
	FP	**Inv**	**Len**	**Ins**	**Shop**	**AS**	**Trans**	**Cons**	**Pres**	**Pers**	**Fin**	**Asset**	**Part**	**Din**	**Trav**	**Hous**	**Well**	**Lit**	**Aud**	**Cont**	**Voice**	**Rep**	**AE**	**AA**
ChatGPT-5.4																								
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Qwen3.5																								
MiniMax M2.7																								
Kimi K2.6																								

References

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